

20260519EB_MN (NCT, ATP, 51)

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2026年5月19日 星期二

split iPSCs into a full 6-well plate on the Friday before differentiation

differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates

keep some iPSCs

2026年5月21日 星期四

T2 -> T3

2026年5月23日 星期六

T3 -> T4 (~14 mL for over-weekend)

2026年5月26日 星期二

T4 -> T5

2026年5月28日 星期四

T5 -> T6

2026年5月30日 星期六

T6 -> T6 (~14mL for over-weekend)

2026年6月2日 星期二

T6 -> T6 (10mL)

coat PLO

2026年6月3日 星期三

T6 -> T7

Dissociation

1. Coat laminin for at least 1h
2. Collect EBs into 50mL tubes (each plate). Wash the plate with 5mL PBS and added them into the tubes. Let EBs settle with gravity.
3. Remove sup. and add 3mL 0.25% trypsin for each (prepare 8mL 0.25% trypsin with 220U DNaseI)
4. Pipette ~8-10 with P1000 and then incubate the tubes in 37C.
5. pipette*10 at 5min, 10min.
6. Stop reaction at 15min by adding 8mL FBS into 50mL tube -> pipette with 5mL pipet ~ 10 times
7. add 8mL CTWM -> pipette ~ 10 times

8. filter cells through 40um cell strainer into new tubes.
9. centrifuge at 200xg, 4min

cell info					^
	A	B	C	D	
1		cell	passage	well#	
2	A	WT11	P2	5	
3	B	WT11	P2	4	

Table1			^
	A	B	
1		volume (mL)	
2	trypsin	4	
3	HI-FBS/CTWM	8	

Table3											/
	A	B	C	D	E	F	G	H	I	J	
1		conc. (M/mL)	survival	total volume	tota cell count	batch	well plate	cell count/well (K)	number of wells	Plate	
2	A	1.4	95	5	7	A	24	250	28	I and II	
3	B	1	99	5	5	B	24	250	20	I and II	

Plate I							^
	1	2	3	4	5	6	
A	110, 15uL (20250613 made)					51, 2uL in total	
B	111, 13.5uL (20250613 made) + 5.5uL (20260515 made)					51, 2uL in total	
C	110, 15uL (20260515 made)					51, 2uL in total (accidentally added 1uL of 20260515 111)	
D	111, 19uL (20250613 made)					51, 2uL in total	

Plate II							^
	1	2	3	4	5	6	
A	vLH92, 10uL in total					110, 20260515, 1uL	
B						110, 20260515, 3uL	
C	vLH92, 10uL in total					111, 20260515, 1uL	
D						111, 20260515, 3uL	

- 2026年6月4日 星期四
- T7-> T7
- 2026年6月6日 星期六
- T7 -> T8
- 2026年6月8日 星期一
- 1/2 T8 -> 1/2 T8
- 2026年6月10日 星期三
- 1/2 T8 -> 1/2 T8
- 2026年6月11日 星期四
- Check virus test (Plate II, column 6) with old scope

Table2				^
	A	B	C	
1		110	111	
2	1uL	With 30% (or 20%), 300ms, expression is good (~200 in nucleus). mCherry+ cells were high.	With 30% (or 20%), 300ms, expression is good (~200 in nucleus). mCherry+ cells were high.	
3	3uL	With 30% (or 20%), 300ms, expression is good and brighter than 1uL. mCherry+ cells were high. Cells in BF looked normal.	With 30% (or 20%), 300ms, expression is good and brighter than 1uL. mCherry+ cells were high. Cells in BF looked normal.	

- 2026年6月12日 星期五
- Imaging vLH51
- Check cytoGEM in neurites with Yoshiho
1. Prepare JFX650
- stock:2.5uM

- working:1nM
- a. add 1:2500 into 2.5mM or 0.15mM glucose media
 - I. 2.4uL in 6mL 0mM glucose without pyruvate
 - II. Prepare media with different glucose concentration
 - b. 1nM Halo dye is insufficient to see TDP43 granules. 6nM Halo can see them much clearer

microscope: new spinning disk confocal
100x, oil (0.11micron/pixel)

Plate I_condition						
	1	2	3	4	5	6
A	A	B	C	D	E	1
B	A	B	C	D	E	2
C	A	B	C	D	E	1&2
D	A	B	C	D	E	3

condition			
	A	B	C
1	condition#	gluc conc (mM)	media
2	1	2.5	no pyruvate or lactate
3	2	0.15	no pyruvate or lactate
4	3	0	no pyruvate or lactate

prepare media with different conc. of glucose with pyruvate							
	A	B	C	D	E	F	G
1		diluted from (mM)	targetconc. (mM)	dilution factor	final volume (uL)	added volume (uL)	media volume
2	1	1100	2.5	440	2000	4.5	1995.5
3	2	2.5	0.15	16.6666666667	1000	60.0	940.0
4				total media volume	3.0	mL	

Treat_time_vLH51											
	A	B	C	D	E	F	G	H	I	J	K
1				vLH51							
2	condition#	condition	Halo dye concentration	treat time + Halo_dye	image time	soma	neurite	TDP43			observation after a while
3	1	2.5mM glucose 1h	1nM	13:15	14:15	1-10	1-16		hard to find moving GEM in soma	It's much easier to find GEMs in neurites	
4	2	0.15mM glucose 1h	1nM	13:45	14:45	11-22	17-32		hard to find moving GEM in soma		neurites are beading and broken at 18:48
5	3	0mM glucose 1h	1nM	14:45	15:45	23-29	33-50		extremely hard to find an alive soma with moving GEM	It's a bit harder to find GEMs moving in neurite, compared to condition 1 and 2	neurites are beading, broken at ~18:49
6	1 (C6)	2.5mM glucose 1h	6nM	15:20	16:20	30-39	51-65	1-8	treated with 6nM Halo dye. I can finally see TDP43 granules		
7	2(C6)	0.15mM glucose 1h	5nM	17:10	18:10		66-80	9-22	neurites looked ok, but some of them are beading		
8	1(A6)	2.5mM glucose 1h+10nM Halo dye	10nM	17:10	48:00				neurites were beading and looked very unhealthy around 18:10. Did not take images		

vLH51 and TDP43 movie_O.C. , 60X, 600x900, 16-bit					
	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	488	12	100	10 ms	
3	640	16	10	100 ms	TDP43 movie. I only see granules moving with 6nM Halo dye instead of 1nM we used to use.

vLH51_fileid-conditions					
	A	B	C	D	E
1	image	treatment	treat_time	pyruvate	type
2	soma_vLH51_MN_100x_1xzoom_001	2.5mM gluc	1h	NT	movie
3	soma_vLH51_MN_100x_1xzoom_002	2.5mM gluc	1h	NT	movie
4	soma_vLH51_MN_100x_1xzoom_003	2.5mM gluc	1h	NT	movie
5	soma_vLH51_MN_100x_1xzoom_004	2.5mM gluc	1h	NT	movie
6	soma_vLH51_MN_100x_1xzoom_005	2.5mM gluc	1h	NT	movie
7	soma_vLH51_MN_100x_1xzoom_006	2.5mM gluc	1h	NT	movie
8	soma_vLH51_MN_100x_1xzoom_007	2.5mM gluc	1h	NT	movie
9	soma_vLH51_MN_100x_1xzoom_008	2.5mM gluc	1h	NT	movie
10	soma_vLH51_MN_100x_1xzoom_009	2.5mM gluc	1h	NT	movie
11	soma_vLH51_MN_100x_1xzoom_010	2.5mM gluc	1h	NT	movie
12	soma_vLH51_MN_100x_1xzoom_011	0.15mM gluc	1h	NT	movie
13	soma_vLH51_MN_100x_1xzoom_012	0.15mM gluc	1h	NT	movie
14	soma_vLH51_MN_100x_1xzoom_013	0.15mM gluc	1h	NT	movie
15	soma_vLH51_MN_100x_1xzoom_014	0.15mM gluc	1h	NT	movie
16	soma_vLH51_MN_100x_1xzoom_015	0.15mM gluc	1h	NT	movie
17	soma_vLH51_MN_100x_1xzoom_016	0.15mM gluc	1h	NT	movie
18	soma_vLH51_MN_100x_1xzoom_017	0.15mM gluc	1h	NT	movie
19	soma_vLH51_MN_100x_1xzoom_018	0.15mM gluc	1h	NT	movie
20	soma_vLH51_MN_100x_1xzoom_019	0.15mM gluc	1h	NT	movie
21	soma_vLH51_MN_100x_1xzoom_020	0.15mM gluc	1h	NT	movie
22	soma_vLH51_MN_100x_1xzoom_021	0mM gluc	1h	NT	movie
23	soma_vLH51_MN_100x_1xzoom_022	0mM gluc	1h	NT	movie
24	soma_vLH51_MN_100x_1xzoom_023	0mM gluc	1h	NT	movie
25	soma_vLH51_MN_100x_1xzoom_024	0mM gluc	1h	NT	movie
26	soma_vLH51_MN_100x_1xzoom_025	0mM gluc	1h	NT	movie
27	soma_vLH51_MN_100x_1xzoom_026	0mM gluc	1h	NT	movie
28	soma_vLH51_MN_100x_1xzoom_027	0mM gluc	1h	NT	movie
29	soma_vLH51_MN_100x_1xzoom_028	0mM gluc	1h	NT	movie
30	soma_vLH51_MN_100x_1xzoom_029	0mM gluc	1h	NT	movie
31	soma_vLH51_MN_100x_1xzoom_030	2.5mM gluc	1h	NT	movie
32	soma_vLH51_MN_100x_1xzoom_031	2.5mM gluc	1h	NT	movie
33	soma_vLH51_MN_100x_1xzoom_032	2.5mM gluc	1h	NT	movie

34	soma_vLH51_MN_100x_1xzoom_033	2.5mM gluc	1h	NT	movie
35	soma_vLH51_MN_100x_1xzoom_034	2.5mM gluc	1h	NT	movie
36	soma_vLH51_MN_100x_1xzoom_035	2.5mM gluc	1h	NT	movie
37	soma_vLH51_MN_100x_1xzoom_036	2.5mM gluc	1h	NT	movie
38	soma_vLH51_MN_100x_1xzoom_037	2.5mM gluc	1h	NT	movie
39	soma_vLH51_MN_100x_1xzoom_038	2.5mM gluc	1h	NT	movie
40	soma_vLH51_MN_100x_1xzoom_039	2.5mM gluc	1h	NT	movie
41	neurite_vLH51_MN_100x_1xzoom_001	2.5mM gluc	1h	NT	movie
42	neurite_vLH51_MN_100x_1xzoom_002	2.5mM gluc	1h	NT	movie
43	neurite_vLH51_MN_100x_1xzoom_003	2.5mM gluc	1h	NT	movie
44	neurite_vLH51_MN_100x_1xzoom_004	2.5mM gluc	1h	NT	movie
45	neurite_vLH51_MN_100x_1xzoom_005	2.5mM gluc	1h	NT	movie
46	neurite_vLH51_MN_100x_1xzoom_006	2.5mM gluc	1h	NT	movie
47	neurite_vLH51_MN_100x_1xzoom_007	2.5mM gluc	1h	NT	movie
48	neurite_vLH51_MN_100x_1xzoom_008	2.5mM gluc	1h	NT	movie
49	neurite_vLH51_MN_100x_1xzoom_009	2.5mM gluc	1h	NT	movie
50	neurite_vLH51_MN_100x_1xzoom_010	2.5mM gluc	1h	NT	movie
51	neurite_vLH51_MN_100x_1xzoom_011	2.5mM gluc	1h	NT	movie
52	neurite_vLH51_MN_100x_1xzoom_012	2.5mM gluc	1h	NT	movie
53	neurite_vLH51_MN_100x_1xzoom_013	2.5mM gluc	1h	NT	movie
54	neurite_vLH51_MN_100x_1xzoom_014	2.5mM gluc	1h	NT	movie
55	neurite_vLH51_MN_100x_1xzoom_015	2.5mM gluc	1h	NT	movie
56	neurite_vLH51_MN_100x_1xzoom_016	2.5mM gluc	1h	NT	movie
57	neurite_vLH51_MN_100x_1xzoom_017	0.15mM gluc	1h	NT	movie
58	neurite_vLH51_MN_100x_1xzoom_018	0.15mM gluc	1h	NT	movie
59	neurite_vLH51_MN_100x_1xzoom_019	0.15mM gluc	1h	NT	movie
60	neurite_vLH51_MN_100x_1xzoom_020	0.15mM gluc	1h	NT	movie
61	neurite_vLH51_MN_100x_1xzoom_021	0.15mM gluc	1h	NT	movie
62	neurite_vLH51_MN_100x_1xzoom_022	0.15mM gluc	1h	NT	movie
63	neurite_vLH51_MN_100x_1xzoom_023	0.15mM gluc	1h	NT	movie
64	neurite_vLH51_MN_100x_1xzoom_024	0.15mM gluc	1h	NT	movie
65	neurite_vLH51_MN_100x_1xzoom_025	0.15mM gluc	1h	NT	movie
66	neurite_vLH51_MN_100x_1xzoom_026	0.15mM gluc	1h	NT	movie
67	neurite_vLH51_MN_100x_1xzoom_027	0.15mM gluc	1h	NT	movie
68	neurite_vLH51_MN_100x_1xzoom_028	0.15mM gluc	1h	NT	movie
69	neurite_vLH51_MN_100x_1xzoom_029	0.15mM gluc	1h	NT	movie
70	neurite_vLH51_MN_100x_1xzoom_030	0.15mM gluc	1h	NT	movie

71	neurite_vLH51_MN_100x_1xzoom_031	0.15mM gluc	1h	NT	movie
72	neurite_vLH51_MN_100x_1xzoom_032	0.15mM gluc	1h	NT	movie
73	neurite_vLH51_MN_100x_1xzoom_033	0mM gluc	1h	NT	movie
74	neurite_vLH51_MN_100x_1xzoom_034	0mM gluc	1h	NT	movie
75	neurite_vLH51_MN_100x_1xzoom_035	0mM gluc	1h	NT	movie
76	neurite_vLH51_MN_100x_1xzoom_036	0mM gluc	1h	NT	movie
77	neurite_vLH51_MN_100x_1xzoom_037	0mM gluc	1h	NT	movie
78	neurite_vLH51_MN_100x_1xzoom_038	0mM gluc	1h	NT	movie
79	neurite_vLH51_MN_100x_1xzoom_039	0mM gluc	1h	NT	movie
80	neurite_vLH51_MN_100x_1xzoom_040	0mM gluc	1h	NT	movie
81	neurite_vLH51_MN_100x_1xzoom_041	0mM gluc	1h	NT	movie
82	neurite_vLH51_MN_100x_1xzoom_042	0mM gluc	1h	NT	movie
83	neurite_vLH51_MN_100x_1xzoom_043	0mM gluc	1h	NT	movie
84	neurite_vLH51_MN_100x_1xzoom_044	0mM gluc	1h	NT	movie
85	neurite_vLH51_MN_100x_1xzoom_045	0mM gluc	1h	NT	movie
86	neurite_vLH51_MN_100x_1xzoom_046	0mM gluc	1h	NT	movie
87	neurite_vLH51_MN_100x_1xzoom_047	0mM gluc	1h	NT	movie
88	neurite_vLH51_MN_100x_1xzoom_048	0mM gluc	1h	NT	movie
89	neurite_vLH51_MN_100x_1xzoom_049	0mM gluc	1h	NT	movie
90	neurite_vLH51_MN_100x_1xzoom_050	0mM gluc	1h	NT	movie
91	neurite_vLH51_MN_100x_1xzoom_051	2.5mM gluc	1h	NT	movie
92	neurite_vLH51_MN_100x_1xzoom_052	2.5mM gluc	1h	NT	movie
93	neurite_vLH51_MN_100x_1xzoom_053	2.5mM gluc	1h	NT	movie
94	neurite_vLH51_MN_100x_1xzoom_054	2.5mM gluc	1h	NT	movie
95	neurite_vLH51_MN_100x_1xzoom_055	2.5mM gluc	1h	NT	movie
96	neurite_vLH51_MN_100x_1xzoom_056	2.5mM gluc	1h	NT	movie
97	neurite_vLH51_MN_100x_1xzoom_057	2.5mM gluc	1h	NT	movie
98	neurite_vLH51_MN_100x_1xzoom_058	2.5mM gluc	1h	NT	movie
99	neurite_vLH51_MN_100x_1xzoom_059	2.5mM gluc	1h	NT	movie
100	neurite_vLH51_MN_100x_1xzoom_060	2.5mM gluc	1h	NT	movie
101	neurite_vLH51_MN_100x_1xzoom_061	2.5mM gluc	1h	NT	movie
102	neurite_vLH51_MN_100x_1xzoom_062	2.5mM gluc	1h	NT	movie
103	neurite_vLH51_MN_100x_1xzoom_063	2.5mM gluc	1h	NT	movie
104	neurite_vLH51_MN_100x_1xzoom_064	2.5mM gluc	1h	NT	movie
105	neurite_vLH51_MN_100x_1xzoom_065	2.5mM gluc	1h	NT	movie
106	neurite_vLH51_MN_100x_1xzoom_066	0.15mM gluc	1h	NT	movie
107	neurite_vLH51_MN_100x_1xzoom_067	0.15mM gluc	1h	NT	movie

108	neurite_vLH51_MN_100x_1xzoom_068	0.15mM gluc	1h	NT	movie
109	neurite_vLH51_MN_100x_1xzoom_069	0.15mM gluc	1h	NT	movie
110	neurite_vLH51_MN_100x_1xzoom_070	0.15mM gluc	1h	NT	movie
111	neurite_vLH51_MN_100x_1xzoom_071	0.15mM gluc	1h	NT	movie
112	neurite_vLH51_MN_100x_1xzoom_072	0.15mM gluc	1h	NT	movie
113	neurite_vLH51_MN_100x_1xzoom_073	0.15mM gluc	1h	NT	movie
114	neurite_vLH51_MN_100x_1xzoom_074	0.15mM gluc	1h	NT	movie
115	neurite_vLH51_MN_100x_1xzoom_075	0.15mM gluc	1h	NT	movie
116	neurite_vLH51_MN_100x_1xzoom_076	0.15mM gluc	1h	NT	movie
117	neurite_vLH51_MN_100x_1xzoom_077	0.15mM gluc	1h	NT	movie
118	neurite_vLH51_MN_100x_1xzoom_078	0.15mM gluc	1h	NT	movie
119	neurite_vLH51_MN_100x_1xzoom_079	0.15mM gluc	1h	NT	movie
120	neurite_vLH51_MN_100x_1xzoom_080	0.15mM gluc	1h	NT	movie

TDP43Halo movie				
	A	B	C	D
1	image	treatment	treat_time	type
2	neurite_TDP43Halo_MN_100x_1xzoom_001	2.5mM gluc	1h	movie
3	neurite_TDP43Halo_MN_100x_1xzoom_002	2.5mM gluc	1h	movie
4	neurite_TDP43Halo_MN_100x_1xzoom_003	2.5mM gluc	1h	movie
5	neurite_TDP43Halo_MN_100x_1xzoom_004	2.5mM gluc	1h	movie
6	neurite_TDP43Halo_MN_100x_1xzoom_005	2.5mM gluc	1h	movie
7	neurite_TDP43Halo_MN_100x_1xzoom_006	2.5mM gluc	1h	movie
8	neurite_TDP43Halo_MN_100x_1xzoom_007	2.5mM gluc	1h	movie
9	neurite_TDP43Halo_MN_100x_1xzoom_008	2.5mM gluc	1h	movie
10	neurite_TDP43Halo_MN_100x_1xzoom_009	0.15mM gluc	1h	movie
11	neurite_TDP43Halo_MN_100x_1xzoom_010	0.15mM gluc	1h	movie
12	neurite_TDP43Halo_MN_100x_1xzoom_011	0.15mM gluc	1h	movie
13	neurite_TDP43Halo_MN_100x_1xzoom_012	0.15mM gluc	1h	movie
14	neurite_TDP43Halo_MN_100x_1xzoom_013	0.15mM gluc	1h	movie
15	neurite_TDP43Halo_MN_100x_1xzoom_014	0.15mM gluc	1h	movie
16	neurite_TDP43Halo_MN_100x_1xzoom_015	0.15mM gluc	1h	movie
17	neurite_TDP43Halo_MN_100x_1xzoom_016	0.15mM gluc	1h	movie
18	neurite_TDP43Halo_MN_100x_1xzoom_017	0.15mM gluc	1h	movie
19	neurite_TDP43Halo_MN_100x_1xzoom_018	0.15mM gluc	1h	movie
20	neurite_TDP43Halo_MN_100x_1xzoom_019	0.15mM gluc	1h	movie
21	neurite_TDP43Halo_MN_100x_1xzoom_020	0.15mM gluc	1h	movie
22	neurite_TDP43Halo_MN_100x_1xzoom_021	0.15mM gluc	1h	movie
23	neurite_TDP43Halo_MN_100x_1xzoom_022	0.15mM gluc	1h	movie

2026年6月13日 星期六

Imaging NCT (batch A) - MMS treatment

microscope: new spinning disk confocal

100x, oil

SirDNA 1:6000

1. Prepare 2.5mM gluc NB-A with pyruvate
2. Take 10.5mL media and add 1.75uL SirDNA
3. Mix well and aliquot 1mL into 10 eppendorfs (for experiments)
4. Prepare 8 eppendorfs with 99uL media on ice to prepare primary dilution
5. Take out the corresponding amount of media from the tubes for the experiments

6. Add 1uL MMS, mix, and dilute again into media right before the experiment

time table for NCT treatment_MMS_Plate I-BatchA												
	A	B	C	D	E	F	G	H	I	J	K	L
1	NCT (plate I)											
2	condition#		condition	drug	media	pyruvate	imaging order	time	imaging	washout with 2.5mM gluc	note	recovery observation on 6/14
3	110											
4	E	A5	MMS 0mM	NT	2.5mM gluc	1mM	1	13:13	14:13	16:00	position 1 might be out of focus	still many dead cells like pre-imaging condition
5	A	A1	MMS 1.5mM	NT	2.5mM gluc	1mM	3	14:35	15:33	15:50		
6	B	A2	MMS 1mM	NT	2.5mM gluc	1mM	5	15:55	16:54	17:17		
7	C	A3	MMS 0.5mM	NT	2.5mM gluc	1mM	7	17:17	18:15	18:39		
8	D	A4	MMS 0.25mM	NT	2.5mM gluc	1mM	9	18:38	19:35	19:57	position 3 out of focus	
9	111											
10	E	B5	MMS 0mM	NT	2.5mM gluc	1mM	2	13:50	14:50	16:00		still many dead cells like pre-imaging condition
11	A	B1	MMS 1.5mM	NT	2.5mM gluc	1mM	4	15:13	16:15	16:30		
12	B	B2	MMS 1mM	NT	2.5mM gluc	1mM	6	16:37	17:35	17:58		
13	C	B3	MMS 0.5mM	NT	2.5mM gluc	1mM	8	17:57	18:55	19:18		
14	D	B4	MMS 0.25mM	NT	2.5mM gluc	1mM	10	19:17	20:15			

fileID_treatment_NCT_20260613_MN_110,111						
	A	B	C	D	E	F
1	image	reporter	treatment	drug	time	pyruvate
2	com_110_MN_no-glu_50%_1s_30s_60x_0000	110	0mM MMS	NT	1h	yes
3	com_110_MN_no-glu_50%_1s_30s_60x_0001	110	0mM MMS	NT	1h	yes
4	com_110_MN_no-glu_50%_1s_30s_60x_0002	110	1.5mM MMS	NT	1h	yes
5	com_110_MN_no-glu_50%_1s_30s_60x_0003	110	1.5mM MMS	NT	1h	yes
6	com_110_MN_no-glu_50%_1s_30s_60x_0004	110	1mM MMS	NT	1h	yes
7	com_110_MN_no-glu_50%_1s_30s_60x_0005	110	1mM MMS	NT	1h	yes
8	com_110_MN_no-glu_50%_1s_30s_60x_0006	110	0.5mM MMS	NT	1h	yes
9	com_110_MN_no-glu_50%_1s_30s_60x_0007	110	0.5mM MMS	NT	1h	yes
10	com_110_MN_no-glu_50%_1s_30s_60x_0008	110	0.25mM MMS	NT	1h	yes
11	com_110_MN_no-glu_50%_1s_30s_60x_0009	110	0.25mM MMS	NT	1h	yes
12	com_111_MN_no-glu_50%_1s_30s_60x_0000	111	0mM MMS	NT	1h	yes
13	com_111_MN_no-glu_50%_1s_30s_60x_0001	111	0mM MMS	NT	1h	yes
14	com_111_MN_no-glu_50%_1s_30s_60x_0002	111	1.5mM MMS	NT	1h	yes
15	com_111_MN_no-glu_50%_1s_30s_60x_0003	111	1.5mM MMS	NT	1h	yes
16	com_111_MN_no-glu_50%_1s_30s_60x_0004	111	1mM MMS	NT	1h	yes
17	com_111_MN_no-glu_50%_1s_30s_60x_0005	111	1mM MMS	NT	1h	yes
18	com_111_MN_no-glu_50%_1s_30s_60x_0006	111	0.5mM MMS	NT	1h	yes
19	com_111_MN_no-glu_50%_1s_30s_60x_0007	111	0.5mM MMS	NT	1h	yes
20	com_111_MN_no-glu_50%_1s_30s_60x_0008	111	0.25mM MMS	NT	1h	yes
21	com_111_MN_no-glu_50%_1s_30s_60x_0009	111	0.25mM MMS	NT	1h	yes

2026年6月14日 星期日

Imaging vLH92 (batch A&B on plate II) - MMS treatment

- *note: cells looked really sick in general
- Do primary dilution and then do the secondary dilution right before imaging. Imaging from E -> A
1. Prepare 50*1 2.5mM gluc NBA + T8 supp + 1mM pyruvate
 2. Dilute MMS 1uL in 99uL media
 3. Prepare different concentration of MMS
 4. Mix well
 5. Right before doing olaparib experiment
 - a. Dilute olaparib (10mM stock) 1:10 to 1mM (1uL into 9uL media)
 - b. Take 5mL media and add 1uL 1mM olaparib

Plate II condition							^
	1	2	3	4	5	6	
A	A	B	C	D	E		
B	A'	B'	C'	D'	E'		
C	D	C	B	A	E		
D	D'	C'	B'	A'	E'		

condition1									/
	A	B	C	D	E	F	G	H	
1	condition #	MMS	media		condition #	MMS	treatment	media	
2	A	MMS 1.5mM	with 1mM pyruvate		A'	MMS 1.5mM	olaparib 200nM	with 1mM pyruvate	
3	B	MMS 1mM	with 1mM pyruvate		B'	MMS 1mM	olaparib 200nM	with 1mM pyruvate	
4	C	MMS 0.5mM	with 1mM pyruvate		C'	MMS 0.5mM	olaparib 200nM	with 1mM pyruvate	
5	D	MMS 0.25mM	with 1mM pyruvate		D'	MMS 0.25mM	olaparib 200nM	with 1mM pyruvate	
6	E	MMS 0mM	with 1mM pyruvate		E'	MMS 0mM	olaparib 200nM	with 1mM pyruvate	

prepare media with different conc. of glucose with pyruvate1								/
	A	B	C	D	E	F	G	
1	diluted from (mM)	targetconc. (mM)	dilution factor	final volume (uL)	added volume (uL)	media volume	100mM pyruvate (uL)	
2	1100	2.5	440	50000	113.6	49886.4	500	
3			total media volume	50.0	mL			

drug calculation													
	A	B	C	D	E	F	G	H	I	J	K	L	M
1			stock (mM)	1st dilution factor	total volume for 1st dil.	stock volume	media volume for 1st dil.	conc. after 1st dilution (mM)	working (mM)	dilution factor from 1st dil.	total volume (mL)	volume of 1st dil.	volume of media
2	MM S	E							0		1000	0	1000
3		D	11800	100	100	1	99	118	0.25	472	1000	2.1	997.9
4		C	11800	100	100	1	99	118	0.5	236	1000	4.2	995.8
5		B	11800	100	100	1	99	118	1	118	1000	8.5	991.5
6		A	11800	100	100	1	99	118	1.5	78.6666666667	1000	12.7	987.3
7										total	5000	27.5	

time table for Perceval treatment_MMS plate I													
	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Perceval		Plate I										
2	condition#		condition	drug	media	pyruvate	time	imaging	washout with 2.5mM gluc	note	recovery observation on 6/15 with bench microscope in TC	PI imaging time	recovery time
3													
4	E	A5	MMS 0mM	NT	2.5mM gluc	1mM	15:50	16:59			similar to pre-imaging		
5	A	A1	MMS 1.5mM	NT	2.5mM gluc	1mM	16:15	17:18			dead	11:32	~7.5h
6	B	A2	MMS 1mM	NT	2.5mM gluc	1mM	16:35	17:35			dead	11:38	~7h
7	C	A3	MMS 0.5mM	NT	2.5mM gluc	1mM	17:00	18:01			a lot of dead cells	11:44	~6.5h
8	D	A4	MMS 0.25mM	NT	2.5mM gluc	1mM	17:20	18:21			some cells survived, but many dead cells		~6h
9													
10	E'	B5	MMS 0mM	olaparib 200nM	2.5mM gluc	1mM	17:45	18:45			similar to pre-imaging		
11	A'	B1	MMS 1.5mM	olaparib 200nM	2.5mM gluc	1mM	18:05	19:03			similar to pre-imaging	11:53	3.5h
12	B'	B2	MMS 1mM	olaparib 200nM	2.5mM gluc	1mM	18:25	19:27			similar to pre-imaging	11:58	3h
13	C'	B3	MMS 0.5mM	olaparib 200nM	2.5mM gluc	1mM	18:47	19:47			similar to pre-imaging	12:04	2.5h
14	D'	B4	MMS 0.25mM	olaparib 200nM	2.5mM gluc	1mM	19:07	20:08	right after the next treatment		similar to pre-imaging, more dead cells	12:12	2h

time table for Perceval treatment_MMS plate II														
	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Perceval		plate II											
2	condition#		condition	drug	media	pyruvate	time	imaging	washout with 2.5mM gluc	note	recovery observation on 6/15 with bench microscope in TC	recover imaging time	recovery time	
3														
4	E	C 5	MMS 0mM	NT	2.5mM gluc	1mM	19:55	20:55	right after the next treatment		similar to pre-imaging			
5	A	C 4	MMS 1.5mM	NT	2.5mM gluc	1mM	20:15	21:15	right after the next treatment		dead			
6	B	C 3	MMS 1mM	NT	2.5mM gluc	1mM	20:37	21:36	right after the next treatment		dead			
7	C	C 2	MMS 0.5mM	NT	2.5mM gluc	1mM	21:02	22:01	right after the next treatment		a lot of dead cells			
8	D	C 1	MMS 0.25mM	NT	2.5mM gluc	1mM	21:20	22:22	right after the next treatment		some cells survived, but many dead cells			
9											similar to pre-imaging			
10	E'	D 1	MMS 0mM	olaparib 200nM	2.5mM gluc	1mM	21:40	22:46	right after the next treatment		similar to pre-imaging			
11	A'	D 5	MMS 1.5mM	olaparib 200nM	2.5mM gluc	1mM	22:06	23:07	right after the next treatment		similar to pre-imaging			
12	B'	D 4	MMS 1mM	olaparib 200nM	2.5mM gluc	1mM	22:55	23:55	right after the next treatment		similar to pre-imaging			
13	C'	D 3	MMS 0.5mM	olaparib 200nM	2.5mM gluc	1mM	23:15	00:17	right after the next treatment		similar to pre-imaging, more dead cells			
14	D'	D 2	MMS 0.25mM	olaparib 200nM	2.5mM gluc	1mM	23:30	00:33	right after the next treatment					

fileid_treat_20260614_MN_92						
	A	B	C	D	E	F
1	image	treatment	drug	treat_time	pyruvate	replicate
2	vLH92_MN_60x-1xzoom_0000	0mM MMS	NT	1h	yes	A
3	vLH92_MN_60x-1xzoom_0001	1.5mM MMS	NT	1h	yes	A
4	vLH92_MN_60x-1xzoom_0002	1mM MMS	NT	1h	yes	A
5	vLH92_MN_60x-1xzoom_0003	0.5mM MMS	NT	1h	yes	A
6	vLH92_MN_60x-1xzoom_0004	0.25mM MMS	NT	1h	yes	A
7	vLH92_MN_60x-1xzoom_0005	0mM MMS	200nM olaparib	1h	yes	A
8	vLH92_MN_60x-1xzoom_0006	1.5mM MMS	200nM olaparib	1h	yes	A
9	vLH92_MN_60x-1xzoom_0007	1mM MMS	200nM olaparib	1h	yes	A
10	vLH92_MN_60x-1xzoom_0008	0.5mM MMS	200nM olaparib	1h	yes	A
11	vLH92_MN_60x-1xzoom_0009	0.25mM MMS	200nM olaparib	1h	yes	A
12	vLH92_MN_60x-1xzoom_0010	0mM MMS	NT	1h	yes	B
13	vLH92_MN_60x-1xzoom_0011	1.5mM MMS	NT	1h	yes	B
14	vLH92_MN_60x-1xzoom_0012	1mM MMS	NT	1h	yes	B
15	vLH92_MN_60x-1xzoom_0013	0.5mM MMS	NT	1h	yes	B
16	vLH92_MN_60x-1xzoom_0014	0.25mM MMS	NT	1h	yes	B
17	vLH92_MN_60x-1xzoom_0015	0mM MMS	200nM olaparib	1h	yes	B
18	vLH92_MN_60x-1xzoom_0016	1.5mM MMS	200nM olaparib	1h	yes	B
19	vLH92_MN_60x-1xzoom_0017	1mM MMS	200nM olaparib	1h	yes	B
20	vLH92_MN_60x-1xzoom_0018	0.5mM MMS	200nM olaparib	1h	yes	B
21	vLH92_MN_60x-1xzoom_0019	0.25mM MMS	200nM olaparib	1h	yes	B

2026年6月15日 星期一

Imaging NCT: Plate I

Cells looked sick in general, especially D1
C4 looks good.

time table for NCT treatment_MMS_Plate I-BatchB										
	A	B	C	D	E	F	G	H	I	J
1	NCT (plate I)									
2	condition#		condition	drug	media	pyruvate	imaging order	time	imaging	
3										
4	E	C1	MMS 0mM	NT	2.5mM gluc	1mM	1	13:53	14:53	
5	A	C4	MMS 1.5mM	NT	2.5mM gluc	1mM	3	15:15	16:13	
6	B	C3	MMS 1mM	NT	2.5mM gluc	1mM	5	16:38	17:36	
7	C	C2	MMS 0.5mM	NT	2.5mM gluc	1mM	7	18:00	19:02	
8										
9	E	D1	MMS 0mM	NT	2.5mM gluc	1mM	2	14:35	15:33	the signal was not good. Very hard to find good cells expressing mCherry
10	A	D4	MMS 1.5mM	NT	2.5mM gluc	1mM	4	15:55	16:54	
11	B	D3	MMS 1mM	NT	2.5mM gluc	1mM	6	17:17	18:19	
12	C	D2	MMS 0.5mM	NT	2.5mM gluc	1mM	8	18:37	19:40	

fileID_treatment_NCT_20260615_MN_110,111						
	A	B	C	D	E	F
1	image	reporter	treatment	drug	time	pyruvate
2	com_110_MN_no-glu_50%_1s_30s_60x_0000	110	0mM MMS	NT	1h	yes
3	com_110_MN_no-glu_50%_1s_30s_60x_0001	110	0mM MMS	NT	1h	yes
4	com_110_MN_no-glu_50%_1s_30s_60x_0002	110	1.5mM MMS	NT	1h	yes
5	com_110_MN_no-glu_50%_1s_30s_60x_0003	110	1.5mM MMS	NT	1h	yes
6	com_110_MN_no-glu_50%_1s_30s_60x_0004	110	1mM MMS	NT	1h	yes
7	com_110_MN_no-glu_50%_1s_30s_60x_0005	110	1mM MMS	NT	1h	yes
8	com_110_MN_no-glu_50%_1s_30s_60x_0006	110	0.5mM MMS	NT	1h	yes
9	com_110_MN_no-glu_50%_1s_30s_60x_0007	110	0.5mM MMS	NT	1h	yes
10	com_111_MN_no-glu_50%_1s_30s_60x_0000	111	0mM MMS	NT	1h	yes
11	com_111_MN_no-glu_50%_1s_30s_60x_0001	111	0mM MMS	NT	1h	yes
12	com_111_MN_no-glu_50%_1s_30s_60x_0002	111	1.5mM MMS	NT	1h	yes
13	com_111_MN_no-glu_50%_1s_30s_60x_0003	111	1.5mM MMS	NT	1h	yes
14	com_111_MN_no-glu_50%_1s_30s_60x_0004	111	1mM MMS	NT	1h	yes
15	com_111_MN_no-glu_50%_1s_30s_60x_0005	111	1mM MMS	NT	1h	yes
16	com_111_MN_no-glu_50%_1s_30s_60x_0006	111	0.5mM MMS	NT	1h	yes
17	com_111_MN_no-glu_50%_1s_30s_60x_0007	111	0.5mM MMS	NT	1h	yes

Imaging Plate II for viability

propidium iodide (PI) stock: 1mg/mL (1.5mM)
dilution: 1500uM/3uM = 500
Prepare 3uM PI in PBS and add 500uL into the media (500uL) to have 1.5uM PI at around 8:30 pm on 6/15
 $500 \times 20 = 10 \rightarrow 10.5\text{mL} + 21\text{uL } 1.5\text{mM PI}$
Use 20X for imaging